Embedded System

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Course info

- IT4210E (3-0-1-6).
- Wed 06:45 PM – 09:15 PM + 3 labs.
- Pre-requisites
  - Computer Architecture
  - Electronics for IT
  - C programming

Course contents

- Overview of embedded system organization, including hardware (the MCU, memory, peripherals) and software.
- Designing embedded system hardware with basic peripheral devices.
- Develop and simulate embedded software based on assembly and high level languages for basic functions, including peripherals (timer/counter, interrupt…), digital input (switch, button, keypad, …), digital output (LEDs, 7-seg LEDs, LED matrix,…), analog interface (ADC,DAC), and serial communications (UART, SPI, I2C, USB,…).
- Introduction to embedded RTOS.

What's the interesting challenge?

- Embedded system is a very very large field, this course only scratches the surface
- Enhanced engineering skills
  - Working with hardware layer
  - Interfacing with the real physical world
- Hands-on experience
  - Learning and developing on development kit

Syllabus (tentative)

- Chap 1: Introduction to embedded systems
- Chap 2: Embedded system organization
- Chap 3: STM32F4 and basic peripheral interfacing
- Chap 4: Serial communication
- Chap 5: Real-world interfacing
- Chap 6: FreeRTOS
- Chap 7: Arduino

Labs

- Lab 1: Basic peripheral interfacing
- Lab 2: Serial communication
- Lab 3: Embedded OS/Energy-aware embedded system

Course evaluation

- Progress evaluation:
  - Labs
- Final exam: quiz + project-based
  - Teams of 2-3 students
  - Project proposed by team (prefered),
  - or selected from list.
- Some samples:
  - Syllabus - EE319K (utexas.edu)
  - EE319K Lab 10 Videos (utexas.edu)



References

- Books and lecture notes
  - Han-Way Huang, Leo Chartrand 2004, Microcontroller: An Introduction to Software & Hardware Interfacing, Cengage Learning.
  - Embedded Systems syllabus from trường Univ. of Cincinnati (EECE 6017C), Univ. of California, Berkeley (EECS 149), Univ. of Pennsylvania (ESE 350), Univ. of Kansas (EECS388)
- Technical documents and tools
  - Datasheet from chip maker: Atmel, Keil…
  - Internet
  - Arduino IDE
  - STM32Cube IDE

Chapter 1: Introduction

Content

- What is embedded system
- Embedded application
- Charateristics of embedded systems

Definition

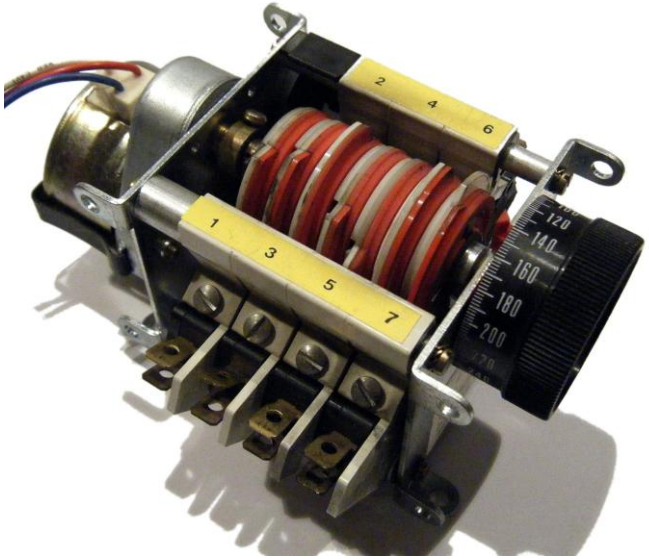
“Dortmund“ Definition: **[Peter Marwedel]**

Embedded systems are information processing systems embedded into a larger product.

Wikipedia:
An embedded system is a computer system with a dedicated function within a larger mechanical or electrical system, often with real-time computing constraints. It is embedded as part of a complete device often including hardware and mechanical parts.

A sample

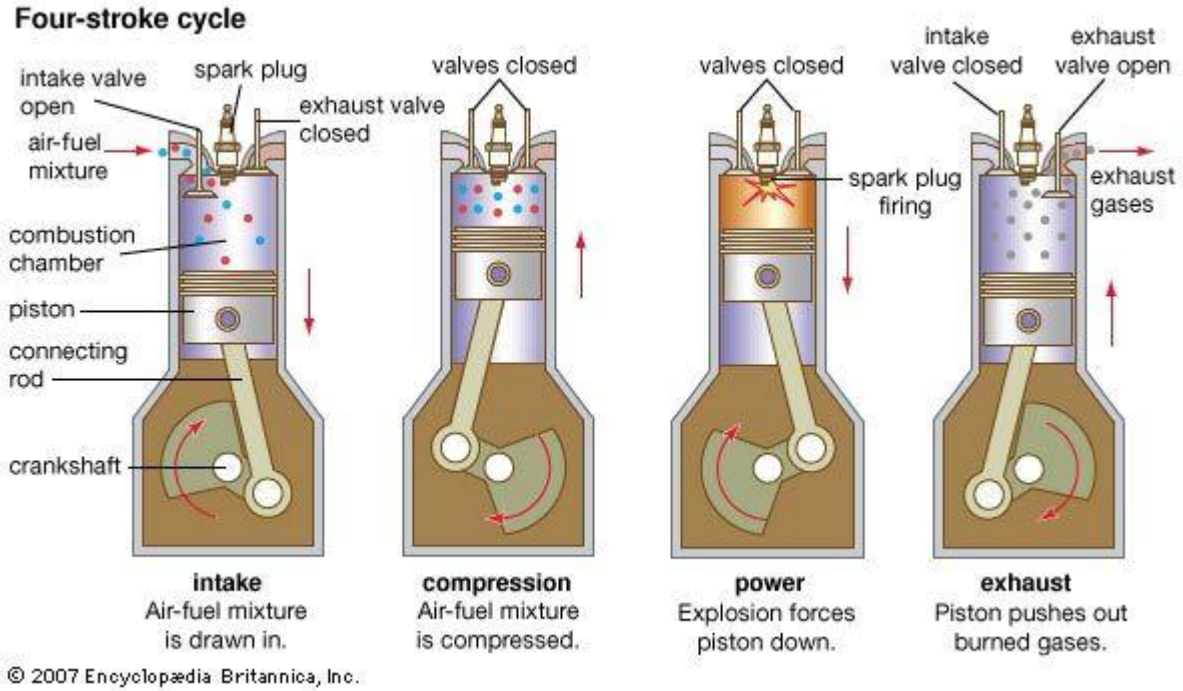








Another sample: ignition in combustion engine





Mechanical ignition (distributor)



Benefit of embedded system

- Before embedded system
  - Logic circuit, hard-wired
  - Electro-mechanical control system
  - Hydraulic control system
  - …
- Benefit of embedding and embedded system:
  - sophisticated control, precise timing, low unit cost, low development cost, high flexibility, small size, and low weight
- System/final product will have improved
  - Performance
  - Features
  - Cost
  - Dependability

Example

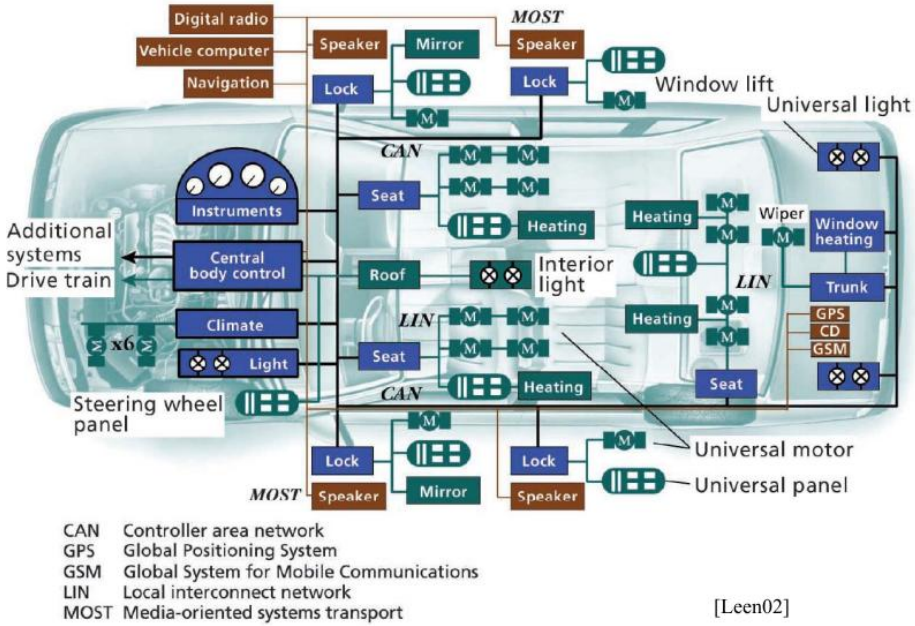












CAN Controller area network
GPS Global Positioning System
GSM Global System for Mobile Communications
LIN Local interconnect network
MOST Media-oriented systems transport

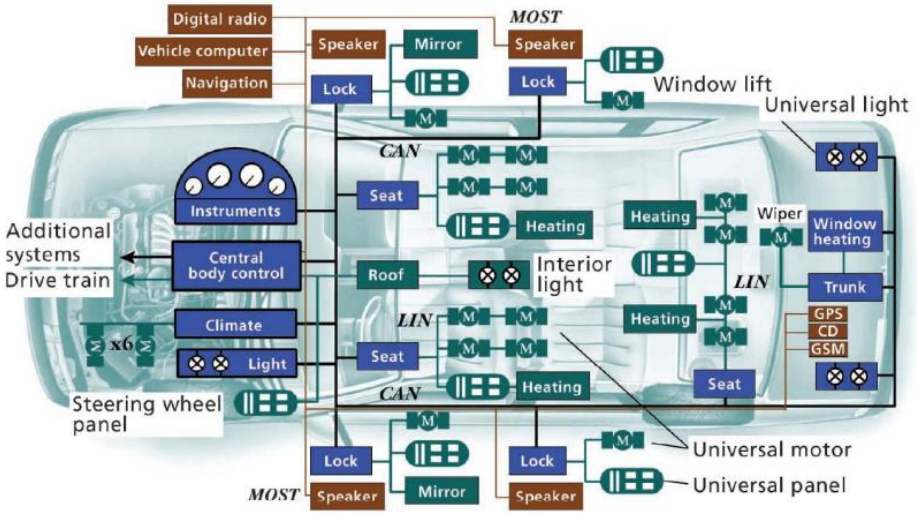
[Leen02]

Automotive electronics

- ABS: Anti-lock braking systems
- ESP: Electronic stability control
- Airbags
- Automatic gearboxes
- Smart keys
- Blind-angle alert systems
- …



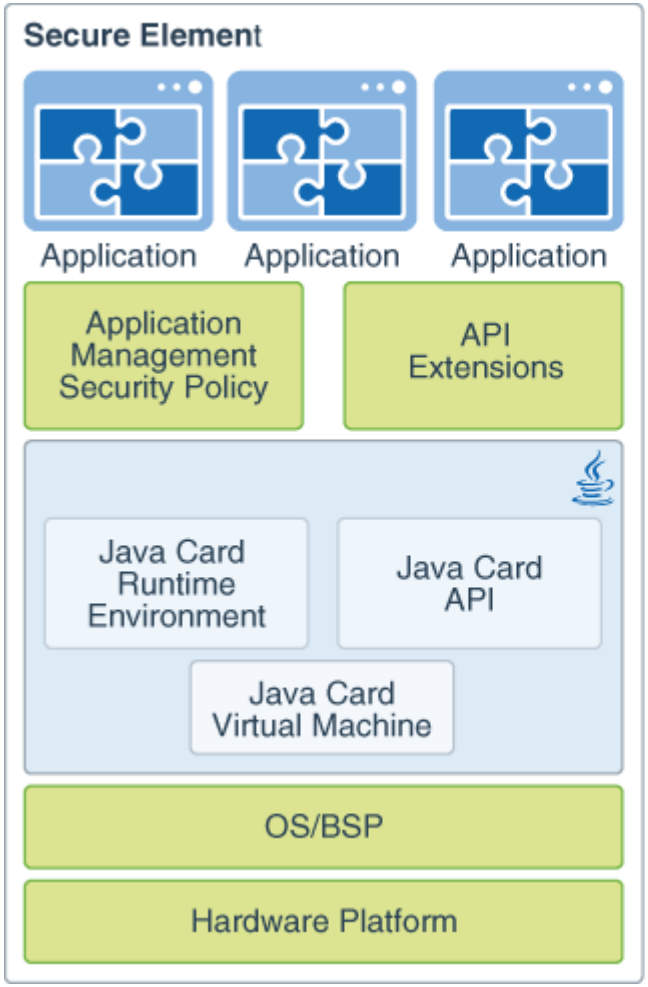
- Up to 100 networked processors



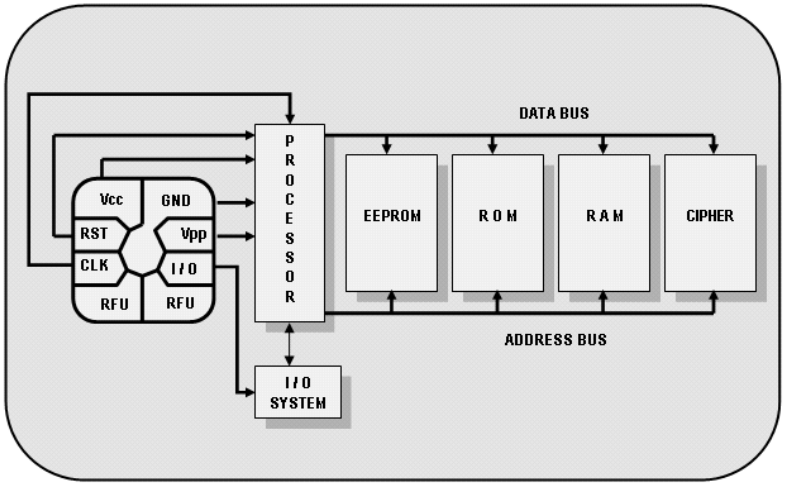
CAN Controller area network
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[Leen02]

Smart card (vd: SIM card)









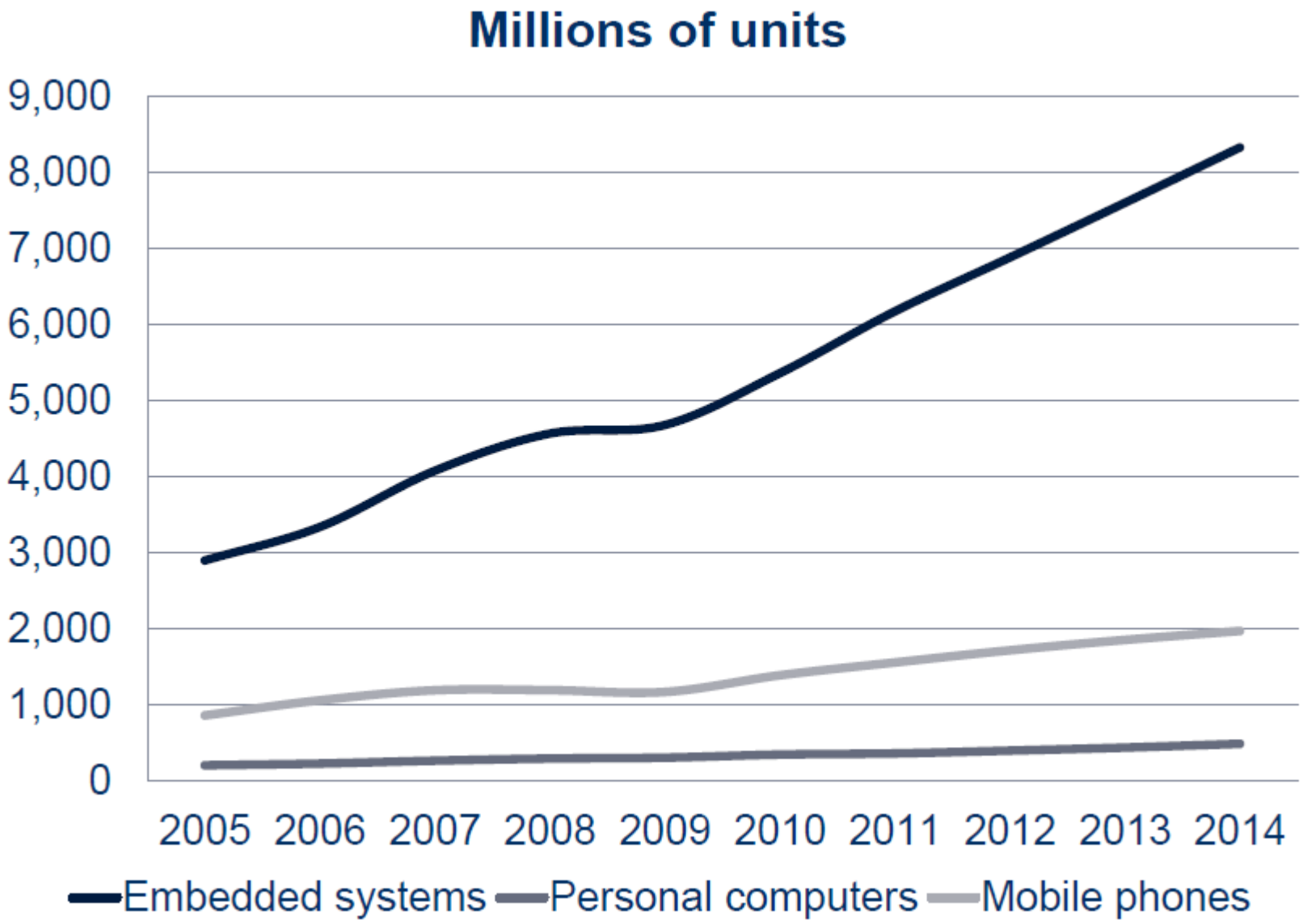
Logistic

- Object tagging : RFID, barcode…
- Object tracking.

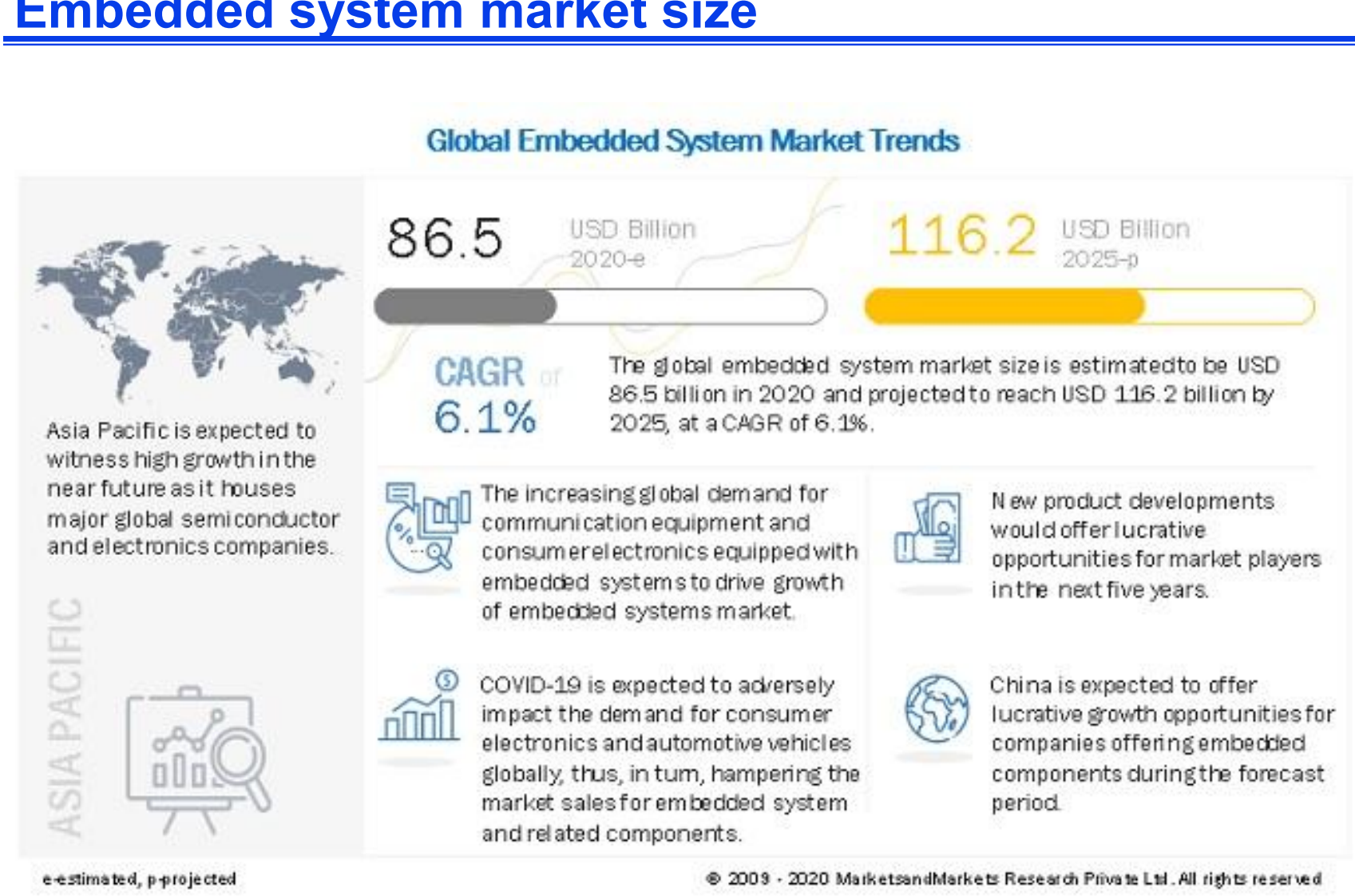


Location	Date	Local Time	Activity
TOKYO, JP	09/03/2016	3:48 P.M.	Delivered
Tokyo, Japan	09/02/2016	7:10 P.M.	Transferred to a local agent for delivery. The delivery will be rescheduled.
	09/02/2016	7:09 P.M.	The delivery change was completed. / The package is being held for a future delivery date.
	09/02/2016	10:07 A.M.	A delivery change for this package is in progress. / We've rescheduled this delivery.
Tokyo, Japan	08/30/2016	7:18 P.M.	The receiver was not available at the time of the final delivery attempt. / We'll contact the sender or receiver about this delivery.
	08/30/2016	2:28 P.M.	The receiver was not available at the time of the final delivery attempt.
Tokyo, Japan	08/29/2016	11:17 A.M.	The receiver was not available at the time of the second delivery attempt. A final attempt will be made.
	08/29/2016	7:48 A.M.	The package is awaiting clearing agency review. / Your package was released by the clearing agency.
Tokyo, Japan	08/26/2016	8:30 P.M.	Transferred to a local agent for delivery. The delivery will be rescheduled.
	08/26/2016	2:15 P.M.	The receiver was not available at the time of the first delivery attempt. A second attempt will be made.
	08/26/2016	1:11 P.M.	Destination Scan
	08/26/2016	1:00 P.M.	Arrival Scan
	08/26/2016	12:55 P.M.	Departure Scan
	08/26/2016	12:38 P.M.	Import Scan
Narita, Japan	08/26/2016	9:03 A.M.	Arrival Scan
	08/26/2016	8:44 A.M.	A late flight has caused a delay. We're adjusting plans and working to deliver your package as quickly as possible.
Anchorage, AK, United States	08/25/2016	8:24 A.M.	Departure Scan
	08/25/2016	6:32 A.M.	Arrival Scan
Ontario, CA, United States	08/25/2016	2:21 A.M.	Departure Scan
Ontario, CA, United States	08/24/2016	11:10 P.M.	Arrival Scan
Commerce City, CO, United States	08/24/2016	9:55 P.M.	Departure Scan
Denver, CO, United States	08/24/2016	8:24 P.M.	Arrival Scan
Boulder, CO, United States	08/24/2016	7:35 P.M.	Departure Scan
	08/24/2016	7:21 P.M.	Origin Scan
United States	08/24/2016	5:03 P.M.	Order Processed: Ready for UPS

Market size of embedded system (IDC)



Embedded system market size

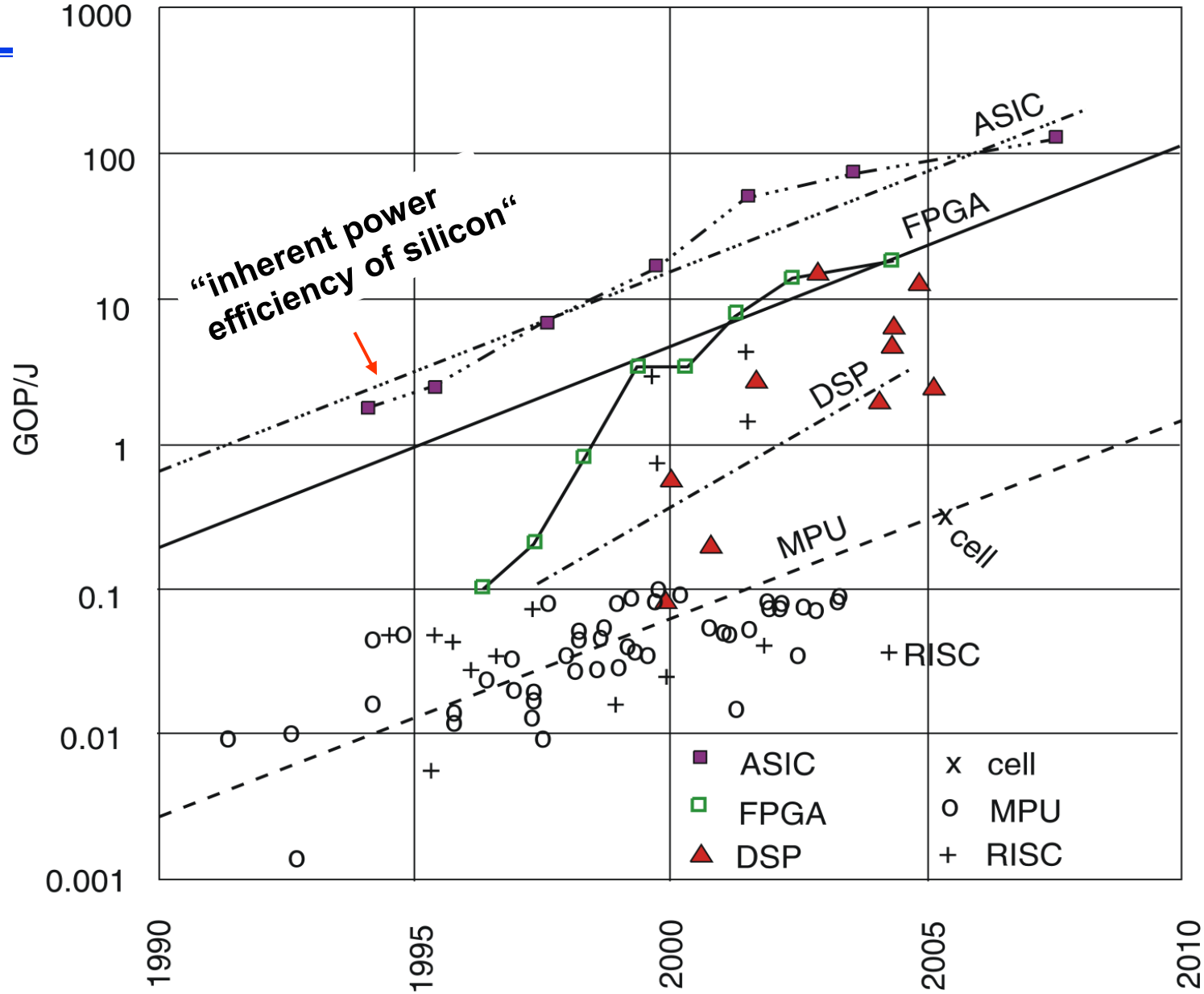


1.2 Embedded systems characteristics

- **Dependability**
  - *Reliability* R(t) = probability of system working correctly provided that is was working at t=0
  - *Maintainability* M(d) = probability of system working correctly d time units after error occurred.
  - *Availability* A(t): probability of system working at time t
  - *Safety*: no harm to be caused
  - *Security*: confidential and authentic communication

Efficiency and resource constraints

- Efficiency
  - Code-size efficient
  - Run-time efficient
  - Weight efficient
  - Cost efficient
  - Energy efficient
- Resource constraints
  - Cost
  - Computing power and storage capacity
  - Physical size
  - Energy consumption

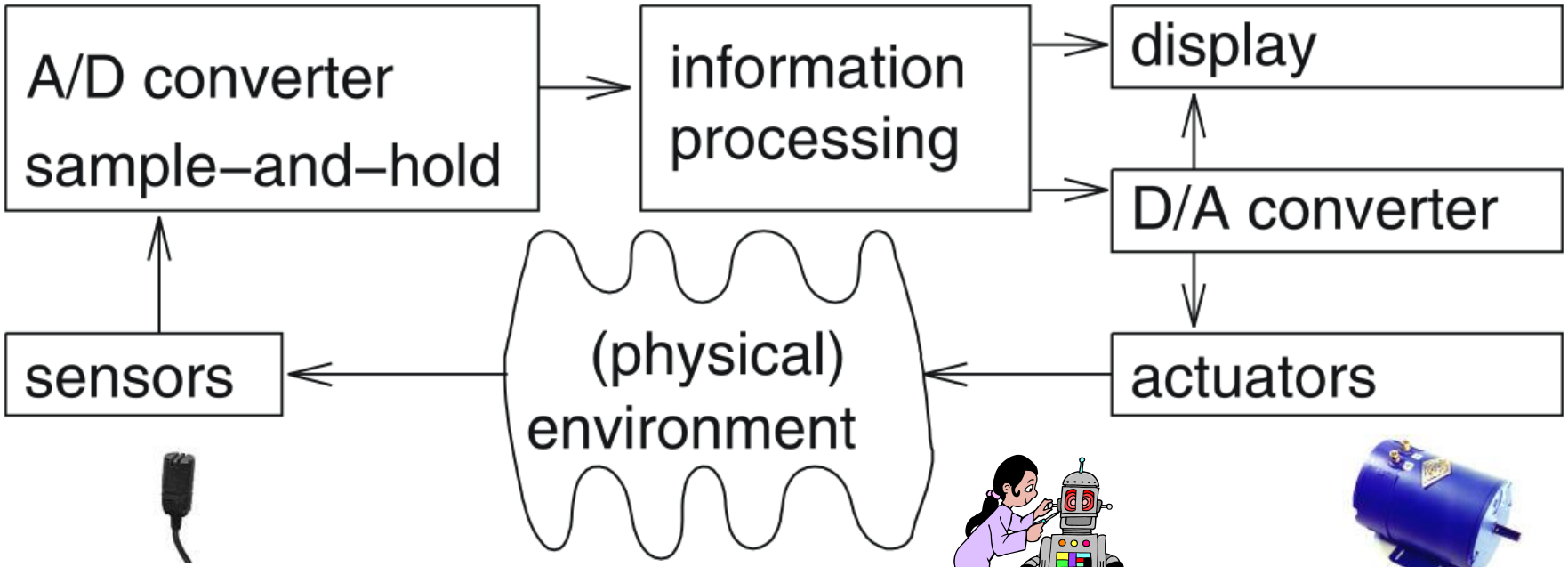


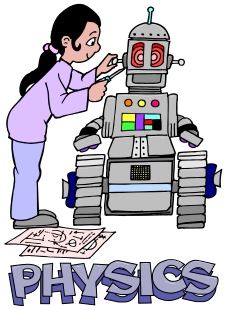
© Hugo De Man, IMEC, Philips, 2007

Embedded System Hardware

- Embedded system hardware is frequently used in a loop (***"hardware in a loop"***):







☞ **cyber-physical systems**

Responsiveness and Real-time constraints

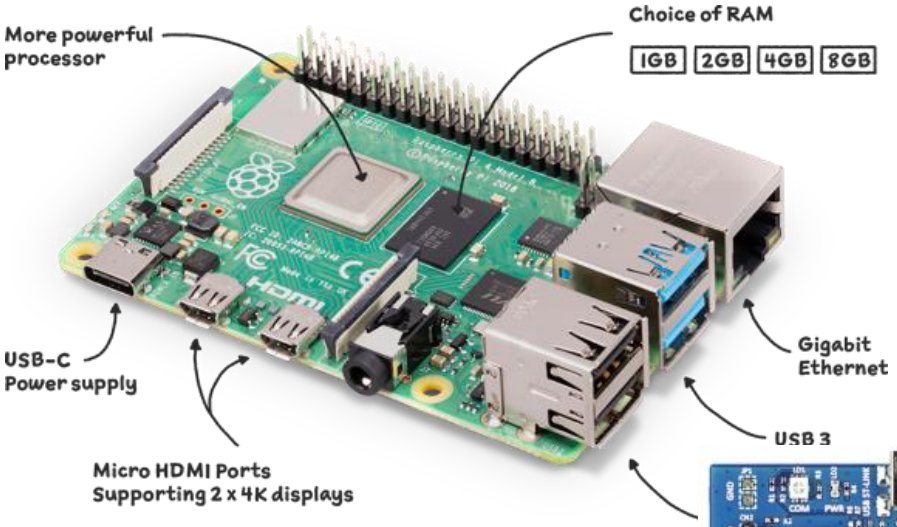
- React to stimuli from the controlled object or external/internal events within the time allowed.
- Many ES must meet **real-time constraints**
  - For real-time systems, right answers arriving too late are wrong.
    - **“A real-time constraint is called hard, if not meeting that constraint could result in a catastrophe“** [Kopetz, 1997].
  - All other time-constraints are called **soft**.
  - A guaranteed system response has to be explained without statistical arguments

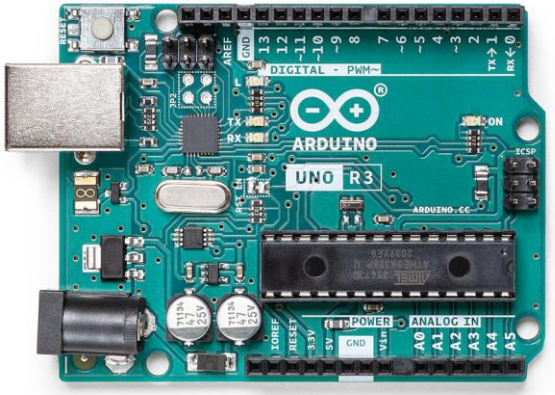
Dedicated systems for special purpose

- **Dedicated** towards a certain **application**
  Knowledge about behavior at design time can be used to minimize resources and to maximize robustness
- **Dedicated user interface**
  (no mouse, keyboard and screen)



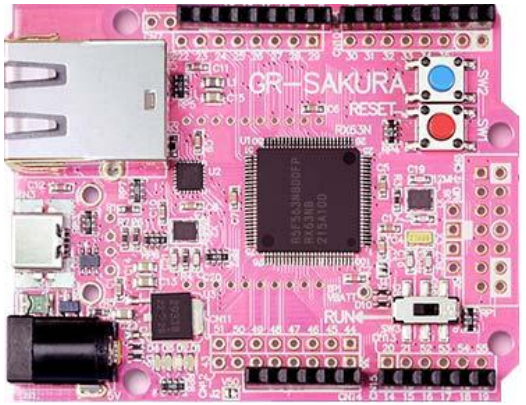
1.3. Embedded computer organization





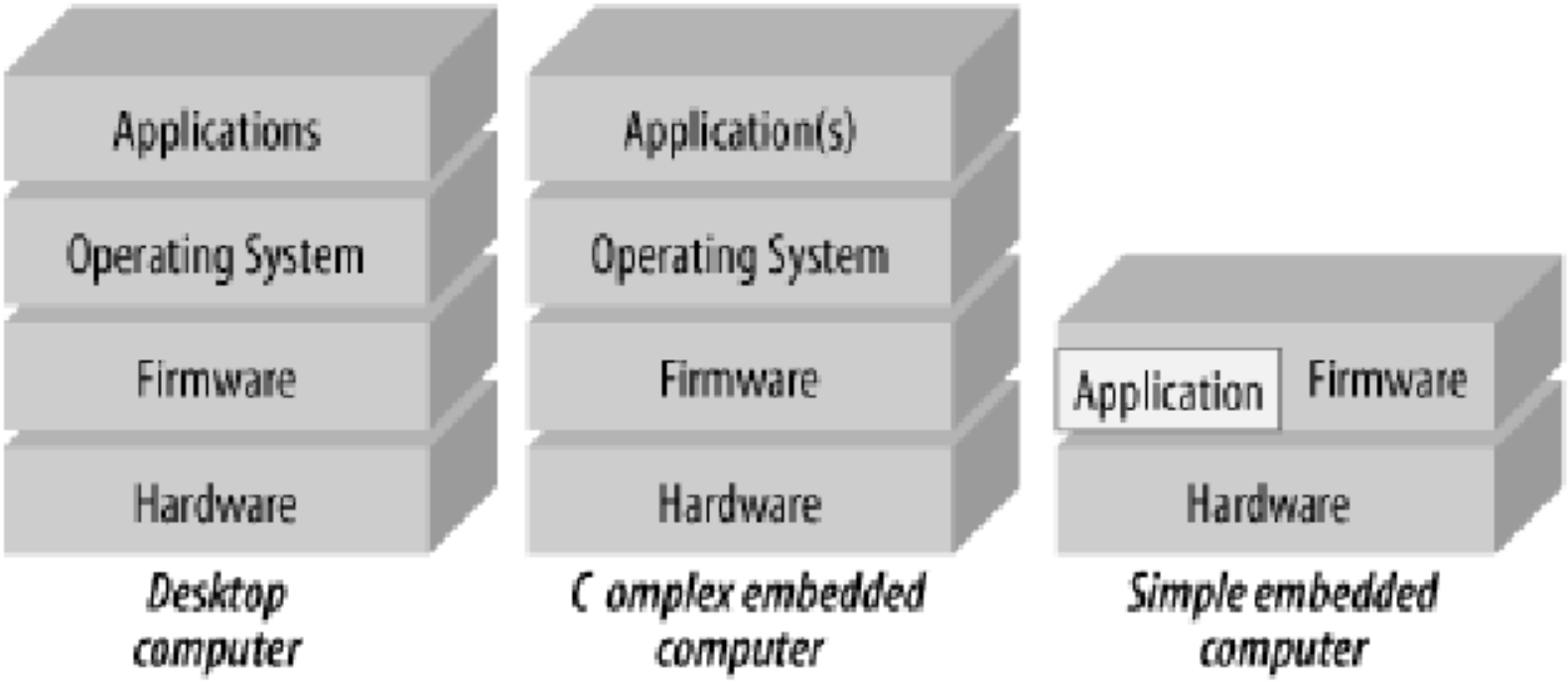






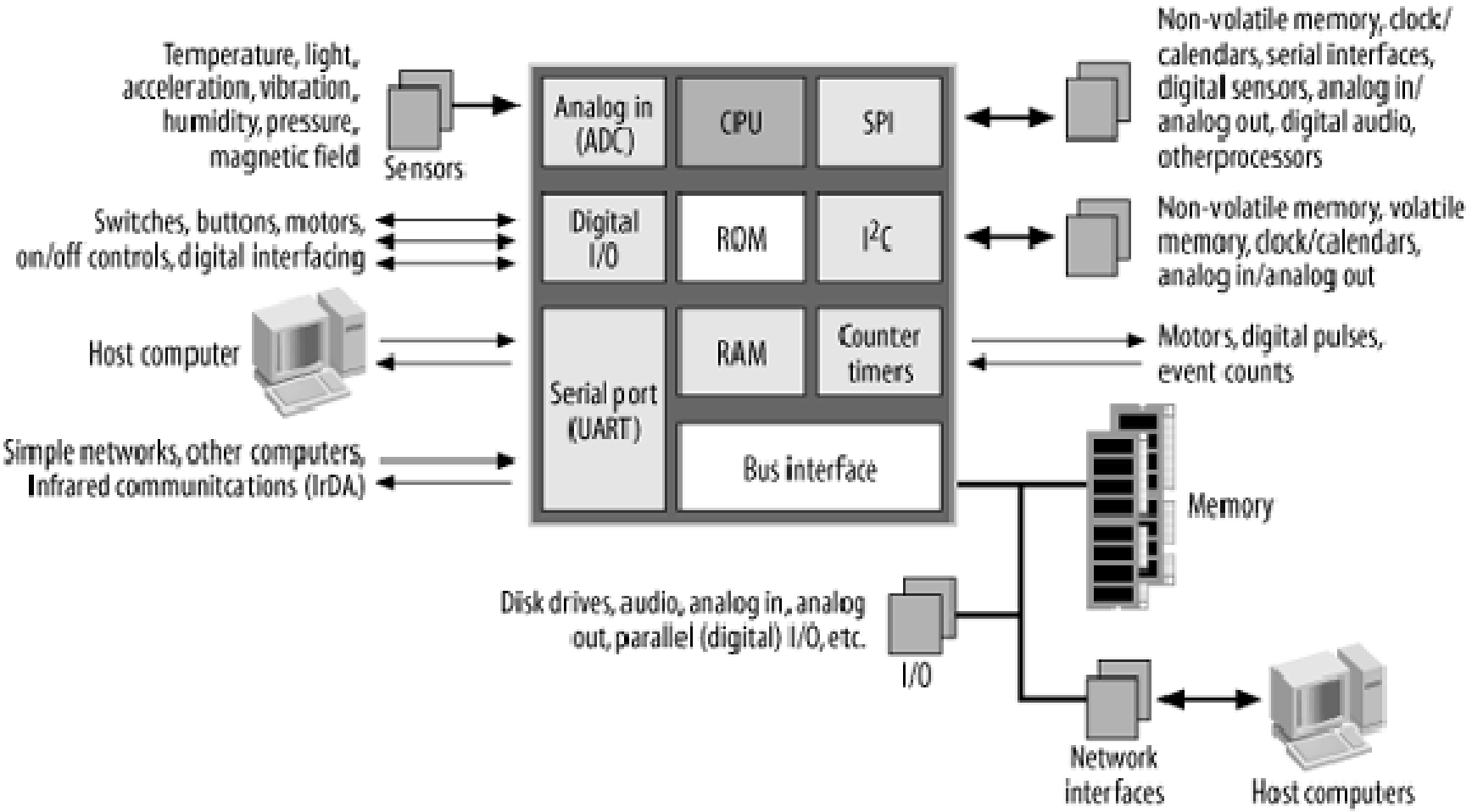
1.3. Embedded computer organization

- System structure



Embedded computer organization

- Central system organization: MCU

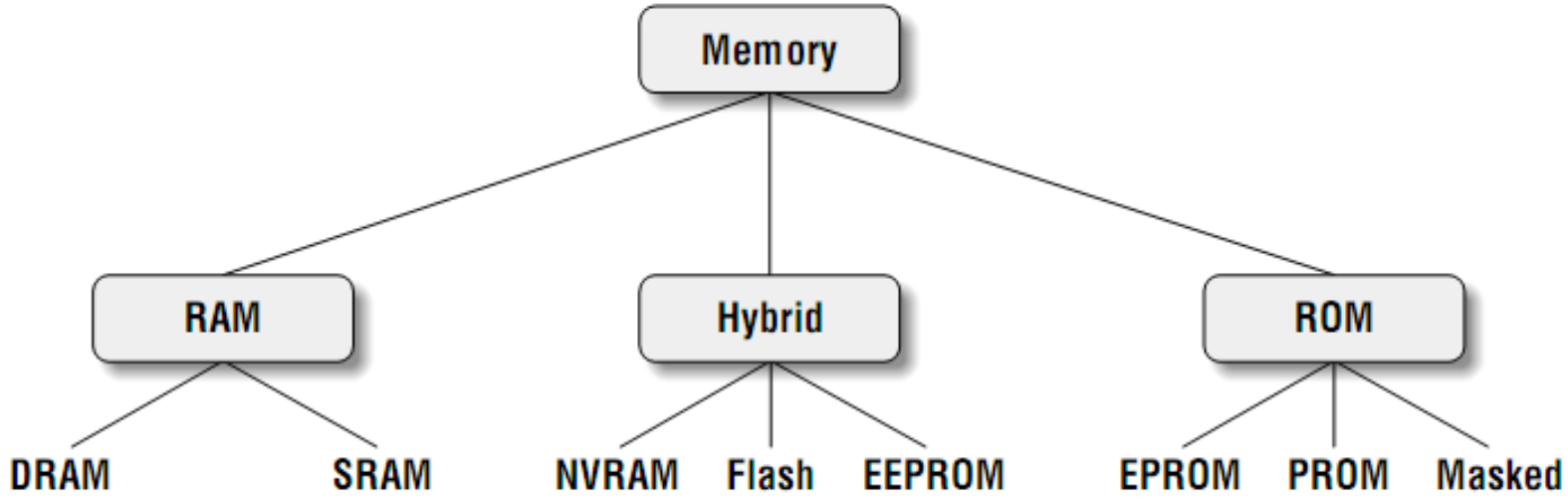


Processing unit in embedded computer

- Microprocessor:
  - CPU packaged as a separate IC.
  - Example: Intel Core, AMD Ryzen,…
- Microcontroller: (Bộ vi điều khiển)
  - A single IC contains basic components of a computer: CPU, Memory (ROM and RAM), basic I/O.
  - Example: AVRs from Atmel, PICs from Microchip, STM32F4 (ARM-based) from STMicroelectronics, …
  - Architecture: 8 bit, 16 bit, 32 bit, (64 bit).
  - On-chip memory or external memory.
- SoC (System-on-chip):
  - High performance components on a single IC: CPU, GPU, modem…
  - Example: Samsung Exynox, Qualcomm Snapdragon, Apple M1…

Memory

- RAM
- ROM, EPROM, EEROM (EEPROM)
- Flash: newest ROM technology, market dominating



Input/Output: peripherals

- Types of peripherals
  - Input: sensors, button, switch, knob, ADC…
  - Output: display, actuators, DAC…
  - Communication: UART, SPI, I2C, CAN,…
  - Storage: EEPROM, Flash memory,…
- I/O programming
  - Programmed I/O (polling)
  - Interrupt-driven I/O
  - Direct Memory Access (DMA)

Exercise

- Read IC datasheet
  - 74HC00
  - DS18B20
- What is the function of each IC
- Footprint of each IC
- How each IC works

End of Chap.1